

Problem 97 (KöMaL Points Contest B. 5223). Define a sequence $\{a_n\}$ by $a_1 = -3$ and $a_{n+1} = a_n + 4 + 4\sqrt{a_n + 4}$. Find a_{2022} .

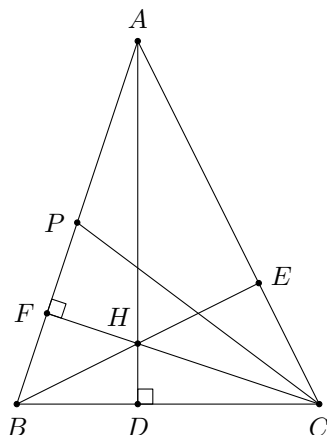
Solution. The answer is 16345845.

We shall prove that $a_n = (2n - 1)^2 - 4$ by induction. The base case $a_1 = -3$ is true. Assuming $a_k = (2k - 1)^2 - 4$, we have

$$a_{k+1} = a_k + 4 + 4\sqrt{a_k + 4} = (2k - 1)^2 + 4\sqrt{(2k - 1)^2} = (2k - 1)(2k + 3) = (2k + 1)^2 - 4.$$

This proves the inductive step. So our claim is true, and $a_{2022} = 4043^2 - 4 = 16345845$. □

Problem 98 (KöMaL Points Contest B. 5193). Let $\triangle ABC$ be an acute triangle with $\angle A = 45^\circ$. Let AD, BE, CF be the altitudes of $\triangle ABC$, and let H be the orthocentre of $\triangle ABC$. Let P be the point on the side AB such that $AP = CH$. Suppose $BD : DC = 2 : 3$. Prove that H is the centroid of $\triangle BCP$.



Solution. Firstly, since $\angle CFA = 90^\circ$ and $\angle FAC = 45^\circ$, we have $AF = CF$. As $AP = CH$, this yields $FP = FH$. Secondly, since $\angle FBH = 90^\circ - \angle BAC = 45^\circ$, we also have $FB = FH$. Combining these, we see that $FP = FB$, which means F is the midpoint of BP .

Let $k = \frac{CH}{HF} = \frac{AP}{PF}$. Applying Menelaus' theorem to $\triangle BCF$ and the line AHD , we obtain

$$1 = \frac{BD}{DC} \times \frac{CH}{HF} \times \frac{FA}{AB} = \frac{2}{3} \cdot k \cdot \frac{k+1}{k+2}.$$

This implies $(k - 2)(2k + 3) = 0$. The only positive root is $k = 2$. Now, as CF is a median of $\triangle CPB$, and H cuts the median in the ratio $2 : 1$, we know that H is the centroid of $\triangle CPB$. □

Problem 99 (KöMaL Points Contest A. 818). Find all positive integers m and n such that each of them divides $9^{|m-n|} + 3^{|m-n|} + 1$.

Solution. The only solutions are $m = n = 1$ and $m = n = 3$. It is easy to check that both satisfy the conditions.

If $m = n$, then $9^{|m-n|} + 3^{|m-n|} + 1 = 3$. The only possibilities are $m = n = 1$ or $m = n = 3$. Therefore, we may assume $m > n$. Let $d = m - n > 0$. Note that $9^{|m-n|} + 3^{|m-n|} + 1 = 9^d + 3^d + 1$ is not divisible by 3. This shows m and n cannot be divisible by 3.

Consider any prime divisor p of m . Let $k = v_3(d)$, and let r be the order of 3 modulo p . Since

$$p \mid m \mid 9^d + 3^d + 1 \mid (3^d - 1)(9^d + 3^d + 1) = 3^{3d} - 1,$$

we have $3^{3d} \equiv 1 \pmod{p}$. This implies $r \mid 3d$. If $r \mid d$, then $3^d \equiv 1 \pmod{p}$, and hence $9^d + 3^d + 1 \equiv 1 + 1 + 1 = 3 \pmod{p}$. This is impossible since $p \mid 9^d + 3^d + 1$ but $p \neq 3$. Therefore, we must have $r \nmid d$. As $r \mid 3d$, we know that $v_3(r) = v_3(3d) = k + 1$. Thus, $3^{k+1} \mid r$.

By the Fermat little theorem, we have $r \mid p-1$. This yields $p \equiv 1 \pmod{3^{k+1}}$. As this holds for every prime divisor of m , we have $m \equiv 1 \pmod{3^{k+1}}$. Similarly, we can deduce $n \equiv 1 \pmod{3^{k+1}}$. This implies $3^{k+1} \mid m-n=d$. However, this is impossible since $v_3(d) = k < k+1$.

Therefore, the only solutions are $m=n=1$ and $m=n=3$. □

Problem 100 (KöMaL Points Contest A. 816). There are 2022 magnetic bars. One end of each magnetic bar is red in colour, while the other end is blue in colour. For some magnetic bars of type 1, the red end has north magnetic polarity and the blue end has south magnetic polarity. For the other magnetic bars of type 2, the red end has south magnetic polarity and the blue end has north magnetic polarity. Each time we can put two magnetic bars together to test whether they are of the same type or of different types. We want to know whether there are equally many type 1 magnetic bars as type 2 magnetic bars. At least how many comparisons are needed so that we are guaranteed to know the answer?

Solution. The answer is 2014. (In general, if 2022 is replaced by $2n$, then the number of comparisons needed is $2n - k$, where k is the number of ones in the binary representation of $2n$.)

Firstly, we show that 2014 comparisons suffices. We consider some groups of magnetic bars. We say that a group is *balanced* if there are equally many type 1 and type 2 magnetic bars, and we say that a group is *polarized* if all magnetic bars are of the same type.

Every time we choose two polarized groups having the same number of magnetic bars. Then we choose a magnetic bar from each group and test whether they are of the same type or not. If they are of the same type, we combine the two groups to form a larger polarized group. If they are of different types, we combine the two groups to form a balanced group. We continue to do this until this cannot be done. At this stage, either all groups are balanced (in which case we are done), or there are some polarized groups of different sizes, possibly together with some balanced groups (which can be ignored as they do not affect the result).

Note that the size of each group must be a power of 2. As $2^m > 2^{m-1} + 2^{m-2} + \dots + 1$, the largest polarized group must contain more magnetic bars than the total of all other polarized groups. This shows the group of all 2022 magnetic bars cannot be balanced, and we are done. Now, as the size of each group is a power of 2, the number of groups remaining cannot be smaller than the number of ones in the binary representation of $2022 = 11111100110_{(2)}$, which is 8. As there are 2022 groups initially, and each comparison reduces the number of groups by 1, we have carried out $2022 - 8 = 2014$ comparisons. This shows 2014 comparisons suffice.

Secondly, we show that 2014 comparisons are needed. Again, we put the magnetic bars into groups. Initially, each magnetic bar forms its own group. When we compare two magnetic bars of different groups, we combine these groups together (but of course the magnetic bars inside the group should be distinguishable). Note that at every moment, we can and can only tell whether every pair of magnetic bars in the same group is of the same type or not (and so it is no use to compare two magnetic bars of the same group). In particular, we know the difference of the numbers of type 1 magnetic bars and type 2 magnetic bars in the same group. We call this the *value* of the group.

At any moment, let $a_1, a_2, \dots, a_k$ be the values of all the groups. We consider some tuples for which every entry is ± 1 . Let S be the number of tuples $(c_1, c_2, \dots, c_k)$ such that

$$c_1 a_1 + c_2 a_2 + \dots + c_k a_k = 0. \quad (1)$$

WLOG assume we compare two magnetic bars from the groups of values a_1 and a_2 respectively. Then the value of the new group formed is either $a_1 + a_2$ or $|a_1 - a_2|$. Let S_1 and S_2 be the number of tuples $(d_1, c_3, \dots, c_k)$ such that

$$d_1(a_1 + a_2) + c_3 a_3 + \dots + c_k a_k = 0, \quad (2)$$

$$d_2 |a_1 - a_2| + c_3 a_3 + \dots + c_k a_k = 0. \quad (3)$$

The solutions to (1) are of two kinds. Those with $c_1 = c_2$ correspond to the solutions to (2) (with $d_1 = c_1$), while those with $c_1 = -c_2$ correspond to the solutions to (3). Therefore, we must have $S = S_1 + S_2$. So at least one of $v_2(S_1) \leq v_2(S)$ and $v_2(S_2) \leq v_2(S)$ must hold. Suppose the result of this comparison is such that $v_2(S)$ does not increase. We show that if this is always the case, then we need at least 2014 comparisons.

Initially, since $a_1 = a_2 = \dots = a_k = 1$, the number of solutions to (1) is $S_0 = \binom{2022}{1011}$. Since $1011 = 1111110011_{(2)}$, by Kummer's theorem, we have $v_2(S_0) = 8$. This shows $v_2(S') \leq 8$ where S' is the number of solutions to (1) when we reach the final stage. At this final stage, if all 2022 magnetic bars form a balanced group, then all groups must be balanced, and hence all their values are 0. Clearly, $S' = 2^k$ as each c_j can be ± 1 . This shows $k \leq 8$, which means at least $2022 - 8 = 2014$ comparisons are needed.

If all 2022 magnetic bars do not form a balanced group, then there is no solution to (1) (otherwise, it could still happen that the group is balanced). This means $S' = 0$, contradicting $v_2(S') \leq 8$. So this cannot happen, and we are done. □